PART B — Practical (Python Programming)

In [38]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats
from scipy.stats import skew, kurtosis
```

In [39]:
```
df = pd.read_csv("C:\\Users\\tanaa\\Desktop\\loan_applications.csv")
df.head()
```

Out[39]:

	Customer_ID	Age	Income	Loan_Amount	Credit_Score	Loan_Term	Default_Status
0	CUST00001	61	77657	105390	735	24	No
1	CUST00002	64	53676	137771	550	12	Yes
2	CUST00003	34	57657	142087	758	84	No
3	CUST00004	55	37802	55708	692	120	Yes
4	CUST00005	31	49745	51943	581	120	No

In [40]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 5200 entries, 0 to 5199
Data columns (total 7 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Customer_ID     5200 non-null   str
 1   Age             5200 non-null   int64
 2   Income          5200 non-null   int64
 3   Loan_Amount     5200 non-null   int64
 4   Credit_Score    5200 non-null   int64
 5   Loan_Term       5200 non-null   int64
 6   Default_Status  5200 non-null   str
```

In [40]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 5200 entries, 0 to 5199
Data columns (total 7 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Customer_ID     5200 non-null   str
 1   Age             5200 non-null   int64
 2   Income          5200 non-null   int64
 3   Loan_Amount     5200 non-null   int64
 4   Credit_Score    5200 non-null   int64
 5   Loan_Term       5200 non-null   int64
 6   Default_Status  5200 non-null   str
```
```
dtypes: int64(5), str(2)
memory usage: 284.5 KB
```

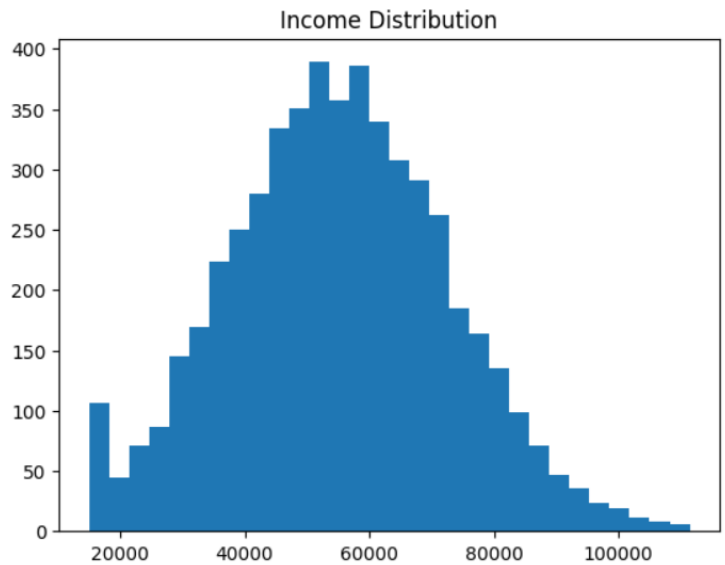
In [26]:
```
print("Mean   :", df['Income'].mean())
print("Median :", df['Income'].median())
print("Mode   :", df['Income'].mode()[0])
```

```
Mean   : 55115.04769230769
Median : 54909.0
Mode   : 15000
```

In [27]:
```
print("Range    :", df['Loan_Amount'].max() - df['Loan_Amount'].min())
print("Variance :", df['Loan_Amount'].var())
print("Std Dev  :", df['Loan_Amount'].std())
```

```
Range    : 398479
Variance : 4947855395.894985
Std Dev  : 70340.99370846977
```

In [41]:
```
plt.hist(df['Income'], bins=30)
plt.title("Income Distribution")
plt.show()
```



In [42]:

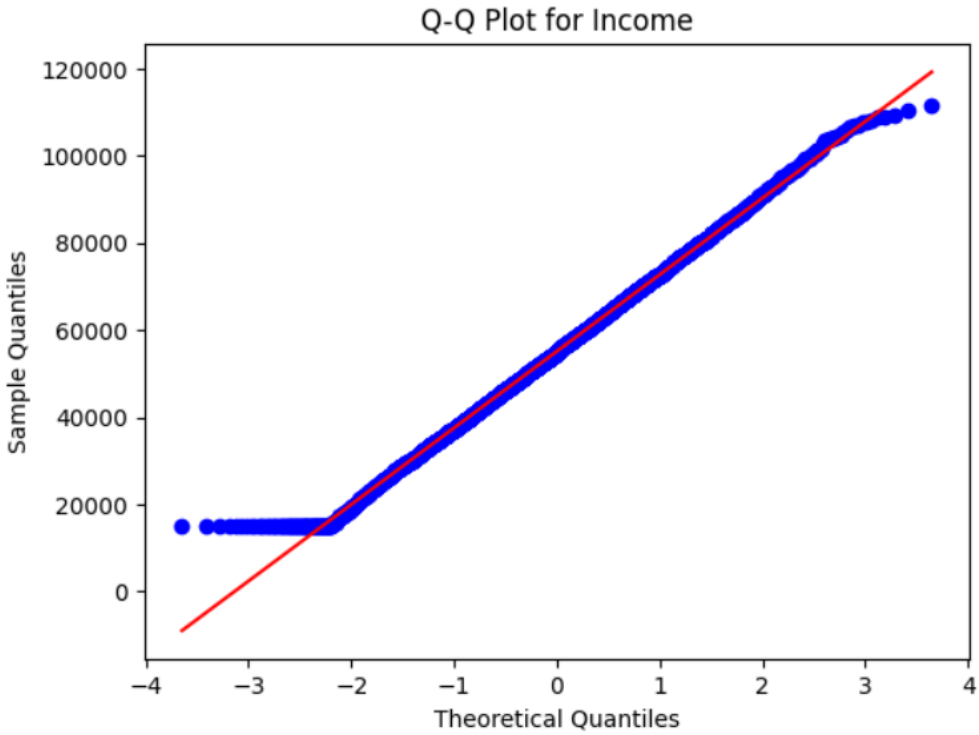
```
total = len(df)
defaults = (df['Default_Status'] == 'Yes').sum()

print("\n--- Default Probability ---")
print("Total:", total)
print("Defaults:", defaults)
print("P(Default):", defaults / total)
```

```
--- Default Probability ---
Total: 5200
Defaults: 1815
P(Default): 0.3490384615384615
```

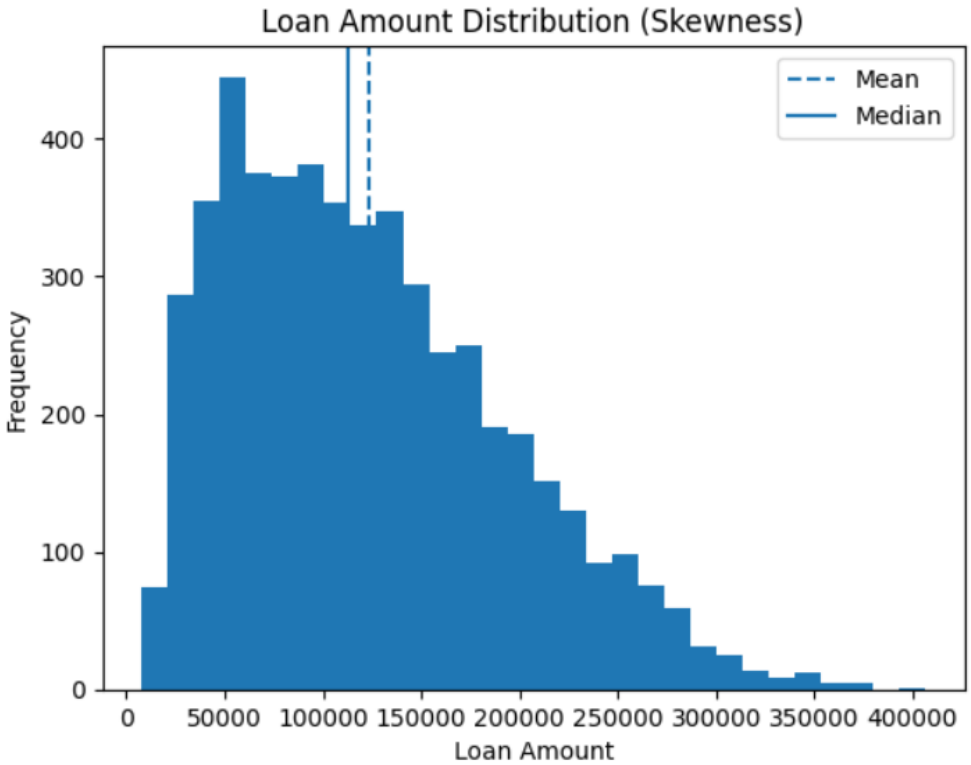
```
P(Default | Low Credit): 0.5404040404040404
P(Default | High Credit): 0.18658280922431866

--- Credit Score ---
Mean: 649.8773076923077
Std Dev: 79.71033310526434

--- Loan Distribution Shape ---
Skewness: 0.6930780383304281
Kurtosis: -0.03859270702073081

--- Vectors ---
v1: [ 77657 105390]
v2: [ 53676 137771]
Dot Product: 18688002822
||v1||: 130910.89240013606
||v2||: 147857.9095517044
Angle (degrees): 15.09873898796453

===== FINAL SUMMARY =====
Income Mean: 55115.04769230769
Default Rate: 0.3490384615384615
Loan Skewness: 0.6930780383304281
Credit Score Mean: 649.8773076923077
Angle: 15.09873898796453
========================
```

In [46]:

```
# Q-Q Plot
stats.probplot(df['Income'], dist='norm', plot=plt)

plt.title("Q-Q Plot for Income")
plt.xlabel("Theoretical Quantiles")
plt.ylabel("Sample Quantiles")
plt.show()
```



In [47]:

```
import matplotlib.pyplot as plt

# Histogram
plt.hist(df['Loan_Amount'], bins=30)

# Mean & Median lines
plt.axvline(df['Loan_Amount'].mean(), linestyle='--', label='Mean')
plt.axvline(df['Loan_Amount'].median(), linestyle='-', label='Median')

plt.title("Loan Amount Distribution (Skewness)")
plt.xlabel("Loan Amount")
plt.ylabel("Frequency")
plt.legend()
```

```
plt.title("Loan Amount Distribution (Skewness)")
plt.xlabel("Loan Amount")
plt.ylabel("Frequency")
plt.legend()

plt.show()
```